AGV / AMR PROCUREMENT RESEARCH REPORT

Industrial indoor material handling — low-profile lifting AMR target
Research consolidated from the supplier/product research completed in this project. Prepared 30 September 2026.

Target requirement

- Indoor factory material handling; low-profile mobile robot similar to the reference style.
- Payload target: ≥300 kg; current preferred target: approximately 600 kg.
- Mecanum / omnidirectional drive preferred.
- Vertical lifting platform/jacking: the top plate physically moves UP/DOWN; not a forklift mast.
- Six wheels were initially preferred. Wheel count must be confirmed by the manufacturer rather than assumed.
- Autonomous navigation preferred, with LiDAR/laser SLAM and obstacle detection.
- China sourcing with potential export to Europe.

1. Key terminology: Laser, LiDAR and Laser SLAM

Laser is the light technology. **LiDAR** is a sensing system that uses laser light to measure distance. **Laser SLAM** combines LiDAR measurements with software that builds a map and estimates the robot's position within that map.

Typical AGV sequence: **LiDAR scans** → **distances are measured** → **SLAM builds/uses a map** → **robot localizes itself** → **route is planned** → **obstacles are detected and avoided.**

Therefore, when a supplier says “Laser SLAM AGV”, it normally means a LiDAR-based autonomous navigation system. The terms are related, but they are not interchangeable: laser is the light source/technology; LiDAR is the distance-sensing system; SLAM is the mapping/localization software method.

2. Main candidate: TIANYUE 600 kg AMR

Item	Publicly verified information
Supplier	TIANYUE / Jintian Robotics
Model / listing	Autonomous Mobile Robot with 600 kg ultra-high load capacity
Payload	600 kg maximum
Drive	Mecanum wheel drive
Lifting	Rear jacking / back lifting device for automatic cargo loading and unloading
Navigation / sensing	3D camera + LiDAR; listing specifies 60 m LiDAR
Battery	LiFePO4
Max speed	1.5 m/s
Obstacle handling	Obstacle detection / avoidance
Customization	Size customizable
MOQ	1 unit
Lead time	30–40 working days
Commercial terms	T/T; wooden-box packaging; 3C/CE stated
Wheel count	Not publicly confirmed for this 600 kg model

Important: Public TIANYUE material confirms 600 kg + Mecanum + rear jacking, but it does **not** confirm whether the chassis has 4 or 6 Mecanum wheels. This is a procurement question that should be answered with a mechanical/3D drawing.

TIANYUE product page: https://www.roboticsagv.com/sale-50544699-autonomous-mobile-robot-with-600kg-ultra-high-load-capacity-and-1-5-m-s-speed-powered-by-lithium-iro.html

TIANYUE supplier page: https://www.roboticsagv.com/supplier-4436044-amr-autonomous-mobile-robot

3. TIANYUE price assessment

The exact public price of the TIANYUE 600 kg Mecanum AMR was not found. The supplier uses quote-based pricing. Comparable Chinese 600 kg jacking AMRs show a broad range depending on drive, sensors, lift, quantity and customization.

Reference	Capacity / features	Observed public price
TZBOT / Made-in-China	600 kg; 60 mm lift; differential; SLAM; 1.5 m/s	US$18,600/sample listing
TZBOT / Alibaba	600 kg; 60 mm lift; SLAM; 1.5 m/s	US$11,800 at 1–199 units on listing
TZBOT category listings	600 kg jacking AMRs	Approx. US$13,900–20,000 FOB on listings
Shenzhen Wellwit	600 kg autonomous jacking platform	Approx. US$17,000 listing
TIANYUE 600 kg	600 kg + Mecanum + rear jacking	Quote required

Budget indication, not a quotation: a preliminary procurement budget of roughly US$10,000–18,000 per unit is reasonable for investigation of a TIANYUE-type 600 kg platform, with customization potentially pushing the price higher. A working planning figure around US$15,000/unit can be used until the factory provides an EXW/FOB quotation.

4. TZBOT 600 kg AMR — the Alibaba model researched

The linked TZBOT model is a **600 kg jacking AMR**. Its public specifications describe Laser SLAM / LiDAR-based navigation, 60 mm lifting, differential drive, approximately 1.5 m/s maximum speed, LiFePO4 battery and a compact chassis around 926 × 756 × 285 mm.

Use of LiDAR: it scans the surroundings and measures distances to walls, machines, pallets and people. SLAM software uses these measurements to create/use a map and determine the robot's position. The LiDAR system is therefore used for autonomous navigation, localization and obstacle detection.

Critical difference from the TIANYUE target: the researched TZBOT 600 kg model is **differential drive**, not the Mecanum configuration preferred for this project.

Alibaba product page: https://www.alibaba.com/product-detail/TZBOT-Smart-AGV-Mobile-Robot-600KG_1601399705959.html

TZBOT 600 kg Made-in-China listing: https://tongzhutech.en.made-in-china.com/product/otXprZTlnKkM/China-Tzbot-China-Manufacturer-600kg-Jacking-up-AMR-Robot-with-Lifting-Device.html

5. Other suppliers / models investigated

Supplier / model	Relevant evidence	Fit to target
TIANYUE TY-AMR-L	300 kg; jacking/back lifting; LiDAR/SLAM; 3D camera; 1.5 m/s	Potential; wheel/drive information is inconsistent across public
TIANYUE TY-CCJS-1000	1000 kg; Mecanum; underbody top lift; 50–100 mm lift stroke	Strong technical match if larger capacity/size is acceptable
TIANYUE TY-FB-JSM-1000/1500	Mecanum + lifting platform; 1000/1500 kg class; industrial/custom	Custom/heavier option
TZBOT TZAMR-SL600	600 kg; 60 mm lift; differential; Laser SLAM	Capacity/lift match; drive does not match Mecanum preference
RoboCT 300 kg platform	300 kg Mecanum mobile platform; 4 Mecanum wheels	Mecanum evidence; integrated lift not confirmed
JXAGV 500B	500 kg Mecanum platform	Mecanum evidence; lifting not combined in same standard mo
JXAGV 500A	500 kg lifting model	Lift evidence; drive is separate/differential rather than verified
Oneway 300JZ	300 kg jacking AGV; 60±2 mm; differential	Lift match; not Mecanum
Riyue custom Mecanum/lifting	Heavy/custom Mecanum + lifting capability	Potential custom source; exact 6-wheel 300+ kg configuration

6. What should be confirmed before buying

- Wheel arrangement: exactly 4 or 6 Mecanum wheels? Request a 3D/mechanical drawing.
- Rated payload versus rated lifting payload: confirm the maximum load that can actually be lifted by the top plate.
- Vertical lift stroke: target approximately 50–100 mm if this matches the transfer application.
- Minimum robot height and total dimensions under the top plate; confirm it can physically enter the required space.
- Top-plate dimensions and allowable center-of-gravity/load distribution.
- Navigation: LiDAR model, scan range, SLAM method, localization accuracy, obstacle detection and safety zones.
- Mecanum omnidirectional performance: lateral movement, turning radius and floor requirements.

- Battery: capacity, charging method, charging time, expected operating hours and battery replacement.
- Safety: emergency stop, safety laser/LiDAR zones, bumper, speed reduction and safety standards.
- Export documents for Europe: CE documentation, applicable declarations, battery UN38.3/MSDS where relevant, electrical documentation and warranty.
- Commercial: EXW and FOB China prices, lead time, spare parts, software/license fees, commissioning and after-sales support.

7. Recommended RFQ message to suppliers

“AGV/AMR for indoor factory material handling. Payload 300–500 kg minimum; preferred 600 kg. Low-profile design similar to our reference. Mecanum omnidirectional drive, preferably 6 wheels. Electric vertical lifting platform/jacking, with the top plate physically moving UP/DOWN, not a forklift mast. Target lifting stroke 50–100 mm and lifting load at least 500–600 kg. LiDAR SLAM navigation, obstacle detection, LiFePO4 battery. Please provide: wheel configuration, 3D/mechanical drawing, lifting mechanism drawing, robot dimensions/minimum height, top plate size, payload and lifting-load rating, LiDAR specification, navigation accuracy, battery data, CE documents, EXW and FOB China price, lead time, warranty and spare-parts pricing.”

8. Procurement conclusion

The **TIANYUE 600 kg AMR** is the closest publicly documented TIANYUE option to the current target because the public specification combines **600 kg payload + Mecanum drive + rear jacking + LiDAR/3D sensing**. The major unresolved requirement is the **wheel count** and exact lifting mechanism/stroke.

The **TZBOT 600 kg** model is useful as a price and technology benchmark: it clearly demonstrates a 600 kg jacking AMR using LiDAR/Laser SLAM, but its publicly documented drive is differential rather than Mecanum.

For purchasing, the next step should therefore be a factory RFQ rather than selecting solely from marketplace specifications. The most important evidence to obtain is the mechanical drawing proving the wheel arrangement and the lifting mechanism, followed by an EXW/FOB quotation.

Key source links

• TIANYUE 600 kg AMR: https://www.roboticsagv.com/sale-50544699-autonomous-mobile-robot-with-600kg-ultra-high-load-capacity-and-1-5-m-s-speed-powered-by-lithium-iro.html

• TIANYUE supplier AMR page: https://www.roboticsagv.com/supplier-4436044-amr-autonomous-mobile-robot

• TIANYUE 600 kg PDF: https://www.roboticsagv.com/doc/50544699/autonomous-mobile-robot-with-600kg-ultra-high-load-capacity-and-1-5-m-s-speed-powered-by-lithium-iro.pdf

• TZBOT Alibaba 600 kg AMR: https://www.alibaba.com/product-detail/TZBOT-Smart-AGV-Mobile-Robot-600KG_1601399705959.html

• TZBOT 600 kg Made-in-China: https://tongzhutech.en.made-in-china.com/product/otXprZTlnKkM/China-Tzbot-China-Manufacturer-600kg-Jacking-up-AMR-Robot-with-Lifting-Device.html

• TIANYUE 1000 kg Mecanum top-lift: https://www.roboticsagv.com/sale-44630074-1000kg-load-agv-automated-guided-vehicle-fully-penetrates-into-jack-up-warehousing.html

Note on evidence: Marketplace specifications and supplier pages can contain inconsistent or incomplete data. Where the public evidence did not verify a requirement (especially wheel count), this report explicitly marks it as unconfirmed rather than assuming it.